

University of Wah

Department of Computer Science

BS Cyber Security_3rd

Lab Manual: DSA LAB 01

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Data Structures

Lab 1

Basic programs in C++, Variable Declarations and Arithmetic Operations &
Selection Statement, Repetition Statement , Arrays

Lab Title:**Basic Programs in C++, Variable Declarations, and Arithmetic Operations**

Lab Objectives:

After completing this lab, you will be able to:

- Understand the structure of a C++ program.
- Declare and initialize different types of variables.
- Perform arithmetic and assignment operations in C++.
- Use basic input/output statements.
- Implement conditional and looping constructs and arrays manipulation.
- Prepare for implementing algorithms in later data structure labs.

Software Requirements:

IDE: Code::Blocks / Dev-C++ / Visual Studio Code (C /C++ Extension)

Compiler: GCC or MinGW C++ Compiler

Operating System: Windows

Introduction:

DSA stands for **Data Structures and Algorithms**. Data structures manage how data is stored and accessed. Algorithms focus on processing this data. Examples of data structures are Array, Linked List, Tree and Heap, and examples of algorithms are Binary Search, Quick Sort and Merge Sort.

Before learning Data Structures, it is essential to understand **basic programming concepts in C++**. C++ provides the foundation for implementing algorithms, manipulating data, and controlling program flow.

A typical C++ program consists of:

- **Header File:** Contains built-in functions (e.g., `#include <iostream>`).
 - **Main Function:** The starting point of every program (`int main()`).
 - **Statements:** Logic or computations written by the programmer.
 - **Output Statement:** Used to display results (`cout`).
 - **Input Statement:** Used to get user input (`cin`).
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2. Structure of a C++ Program

```
#include <iostream>    // Header file
using namespace std;   // Allows use of standard names (cout, cin)

int main() {           // Main function starts here
    cout << "Welcome to C++ Programming!"; // Output statement
    return 0;          // Program ends successfully
}
```

Explanation:

- `#include <iostream>` → for input/output stream.
- `using namespace std;` → allows use of standard library objects.
- `cout` → prints data on the screen.
- `return 0;` → terminates the program.

3. Variable Declarations

Variables are memory locations used to store data temporarily.

Example:

```
#include <iostream>
using namespace std;

int main() {
    int age = 20;        // Integer variable
    float marks = 87.5;  // Float variable
    char grade = 'A';    // Character variable
    string name = "Wasifa"; // String variable

    cout << "Name: " << name << endl;
    cout << "Age: " << age << endl;
    cout << "Marks: " << marks << endl;
    cout << "Grade: " << grade << endl;

    return 0;
}
```

Lab Activity 2:

Declare **five variables** (int, float, char, bool, string), assign values, and print them. Also, print the **size** of each data type using the sizeof() operator.

4. Arithmetic and Assignment Operations

C++ supports standard arithmetic operators:

Operator	Description	Example (a=10, b=3)	Result
+	Addition	a + b	13
-	Subtraction	a - b	7
*	Multiplication	a * b	30
/	Division	a / b	3
%	Modulus (remainder)	a % b	1

```
#include <iostream>
using namespace std;

int main() {
    int a, b;
    cout << "Enter two numbers: ";
    cin >> a >> b;

    cout << "Addition: " << a + b << endl;
    cout << "Subtraction: " << a - b << endl;
    cout << "Multiplication: " << a * b << endl;
    cout << "Division: " << a / b << endl;
    cout << "Modulus: " << a % b << endl;

    return 0;
}
```

Example:

Additional Concept: Increment, Decrement, and Assignment Operators

```
int x = 10;  
x += 5; // same as x = x + 5  
x--;    // same as x = x - 1
```

Lab Activity 3:

Write a program that takes an integer input and displays its **square, cube, and half** using arithmetic operators.

Then use ++ and -- operators to show incremented and decremented values.

5. Conditional Statements (if-else, switch)

```
#include <iostream>  
using namespace std;  
  
int main() {  
    int num;  
    cout << "Enter a number: ";  
    cin >> num;  
  
    if (num > 0)  
        cout << "Positive";  
    else if (num < 0)  
        cout << "Negative";  
    else  
        cout << "Zero";  
  
    return 0;  
}
```

Example:

```
#include <iostream>
using namespace std;

int main() {
    char grade;
    cout << "Enter grade (A-D): ";
    cin >> grade;

    switch(grade) {
        case 'A': cout << "Excellent"; break;
        case 'B': cout << "Good"; break;
        case 'C': cout << "Average"; break;
        case 'D': cout << "Fail"; break;
        default: cout << "Invalid Grade";
    }
    return 0;
}
```

Example: Switch Case

Lab Activity 4:

Write a program that takes an integer input and checks whether it is **even or odd** using if-else, and then print a grade message using a switch-case.

6. Loops in C++

Loops are used to execute a block of code repeatedly until specific condition is true.

(a) For Loop

```
for (int i = 1; i <= 5; i++) {
    cout << i << " ";
}
```

(b) While Loop

```
int i = 1;
while (i <= 5) {
    cout << i << " ";
    i++;
}
```

(c) Do-While Loop

```
int i = 1;
do {
    cout << i << " ";
    i++;
} while (i <= 5);
```

Lab Activity 5:

Use all three types of loops to print numbers from **1 to 10**, then modify the code to print only **even numbers** using `continue`.

Lab Tasks

Task 1:

Write a program that displays your **name, registration number, semester, and the year you joined university**, then calculates how many years you have studied.

Task 2:

Write a program that takes **two integers** and performs:

- Sum, Difference, Product, Quotient, and Remainder
- Also, print which of the two numbers is **greater**, using an if-else condition.

Task 3:

Write a program to calculate the **area and perimeter of a rectangle** using given width and height.

Then, using an if statement, check whether it is a **square or rectangle**.

Task 4:

Write a program that calculates the **average of three numbers**, then checks if the average is **greater than or equal to 50** → print “Pass”, else print “Fail”.

Task 5:

Write a program that:

- Takes three subject marks from the user.
- Calculates **total, average, and percentage**.
- Displays a **grade** using a switch or if-else:
 - $\geq 80 \rightarrow A$
 - $\geq 70 \rightarrow B$
 - $\geq 60 \rightarrow C$
 - $\geq 50 \rightarrow D$
 - Else $\rightarrow$ Fail

Task 6: (Mini Project – Smart Calculator)

Write a program that works as a simple calculator.

- Take two numbers and an operator (+, -, *, /, %) from the user.
- Perform the corresponding arithmetic operation.
- Handle invalid operators gracefully using switch-case.
- Allow user to **repeat the calculation** using a loop until they choose to exit.

Task 7:

Write a program that initializes an array of **N numbers** (where N is entered by the user), and prints:

- The **square** of each element.
- The **sum of all squares** at the end.

Note: Deadline of Lab task is 25 October till 10 pm

